

Artificial Intelligence Exam Notes

CT 710 (BEI, IV-I) · CT 653 (BCT, III-II) · CT 78506 (BEX, IV-II) · TU, IOE · tiers over 41 papers, 2069–2082 BS

6. Machine Learning – Solved Problems

Every problem opens with the question **exactly as the paper prints it**, data table and all, so it can be attempted from this document alone. · **Three ID3 data tables over six papers, three genetic-algorithm runs, and the fuzzy arithmetic.** · Every entropy, gain, fitness value and membership number printed here is recomputed independently by `verify.py`; 242 checks pass across the subject.

Two things to know before you start.

- **The two “play” tables have different answers.** They look the same – fourteen rows, same four attributes, nine Yes and five No. But 79 Jth changes four rows, and that moves the root attribute from **Outlook** to **Windy**. Copying the golf answer into the cricket question loses the whole 7 marks. Problems 1.1 and 1.2 are printed side by side for exactly this reason.
- 75 Bh **prints no chromosomes.** The question sets up the fitness function, says “the initial population consist of four individuals with the following chromosomes” – and then the page runs on to Q7. Problem 2.2 reproduces the defect as printed and works the standard four chromosomes that the textbook version of this question carries.

6.N.1 Decision trees: the method, used identically in all four problems

The recipe

- **Step 1.** Count the classes in the whole set S . Write P (positives) and N (negatives), then $H(S) = -\frac{P}{P+N} \log_2 \frac{P}{P+N} - \frac{N}{P+N} \log_2 \frac{N}{P+N}$.
- **Step 2.** For each attribute, split S by its values and fill *one row per value* of the table below: the counts, that branch's entropy, and the branch weighted by its share of S .
- **Step 3.** Sum the weighted column. That is the **residual entropy** after splitting on that attribute.
- **Step 4.** $\text{Gain} = H(S) - \text{residual}$. Compare the gains; the largest wins.
- **Step 5.** Recurse on each branch that is not already pure.

The table idiom, used unchanged in every problem below

Attribute	Value	P	N	$ S_v $	$H(S_v)$	$\frac{ S_v }{ S } H(S_v)$
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Two logarithms are all you ever need

	Split $P:N$			
$\log_2 x = \frac{\ln x}{\ln 2}$, and these five cover every split in this chapter.	H	1:1	1:2 or 2:1	1:3 or 3:1
		1.0000	0.9183	0.8113

Also $H(1:6) = 0.5917$, $H(2:5) = 0.8631$, $H(5:9) = 0.9403$, $H(3:5) = 0.9544$, and $H = 0$ whenever one side is empty.

6.N.2 Problem 1.1 – the “play golf” table (3 papers)

Select the root attribute of the decision tree from the given sample data 8 2+7

PIN 3/41 74 Bh 2+7, 74 Ma 8, 72 Ash 8 · all three print the same fourteen rows · 74 Bh asks for the ID3 approach first, which is \$6.5

QUESTION AS PRINTED 74 Bh · Q4 · 2+7

State the approach for learning using ID3 and Select the root attribute of the decision-tree from given sample data.

Outlook	Temperature	Humidity	Windy	Play golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Step 1 Entropy of the whole set

14 rows: 9 Yes, 5 No.

$$H(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.4098 + 0.5305 = \mathbf{0.9403}$$

Steps 2 and 3 Split on each attribute in turn

Attribute	Value	P	N	S _v	H(S _v)	$\frac{ S_v }{14} H(S_v)$
Outlook	Rainy	2	3	5	0.9710	$\frac{5}{14}(0.9710) = 0.3468$
	Overcast	4	0	4	0	0
	Sunny	3	2	5	0.9710	$\frac{5}{14}(0.9710) = 0.3468$
residual = 0.6935						Gain = 0.9403 - 0.6935 = 0.2467
Temperature	Hot	2	2	4	1.0000	0.2857
	Mild	4	2	6	0.9183	0.3936
	Cool	3	1	4	0.8113	0.2318
residual = 0.9111						Gain = 0.0292
Humidity	High	3	4	7	0.9852	0.4926
	Normal	6	1	7	0.5917	0.2958
residual = 0.7885						Gain = 0.1518
Windy	False	6	2	8	0.8113	0.4636
	True	3	3	6	1.0000	0.4286
residual = 0.8922						Gain = 0.0481

Step 4 Compare

Attribute	Gain
Outlook	0.2467
Humidity	0.1518
Windy	0.0481
Temperature	0.0292

Answer. The root attribute is **Outlook**, with the highest information gain, **0.2467**.

Why Outlook wins

- Outlook is the only attribute with a **pure branch** – all four **Overcast** days are Yes – so it removes more uncertainty than any other.
- Temperature removes almost none: its three branches are as mixed as the parent.
- **Overcast** (rows 3, 7, 12, 13) is all Yes, so it becomes a **leaf** at once. **Rainy** (2:3) and **Sunny** (3:2) both still have $H = 0.9710$, so each needs its own split: Step 5.

Step 5 Recurse: Steps 1 to 4 again, inside each impure branch (not asked, but it proves the root was right)

Rows numbered 1 to 14 top to bottom. Each branch uses only *its own* five rows and only the three attributes Outlook has not used. Both branches start from $H = 0.9710$, so the residual is the whole story.

Outlook = Rainy rows 1, 2, 8, 9, 11 2 Yes, 3 No

Attribute	Value (rows)	P	N	H	Gain
Humidity	High (1,2,8)	0	3	0	0.9710
	Normal (9,11)	2	0	0	
Temperature	Hot (1,2)	0	2	0	0.5710
	Mild (8,11)	1	1	1.0000	
	Cool (9)	1	0	0	
Windy	False (1,8,9)	1	2	0.9183	0.0200
	True (2,11)	1	1	1.0000	

Humidity: residual 0, gain = $0.9710 - 0 = 0.9710$. Temperature: residual $\frac{2}{5}(1) = 0.4000$, gain 0.5710. Windy: residual $\frac{3}{5}(0.9183) + \frac{2}{5}(1) = 0.9510$, gain 0.0200. **Split on Humidity.**

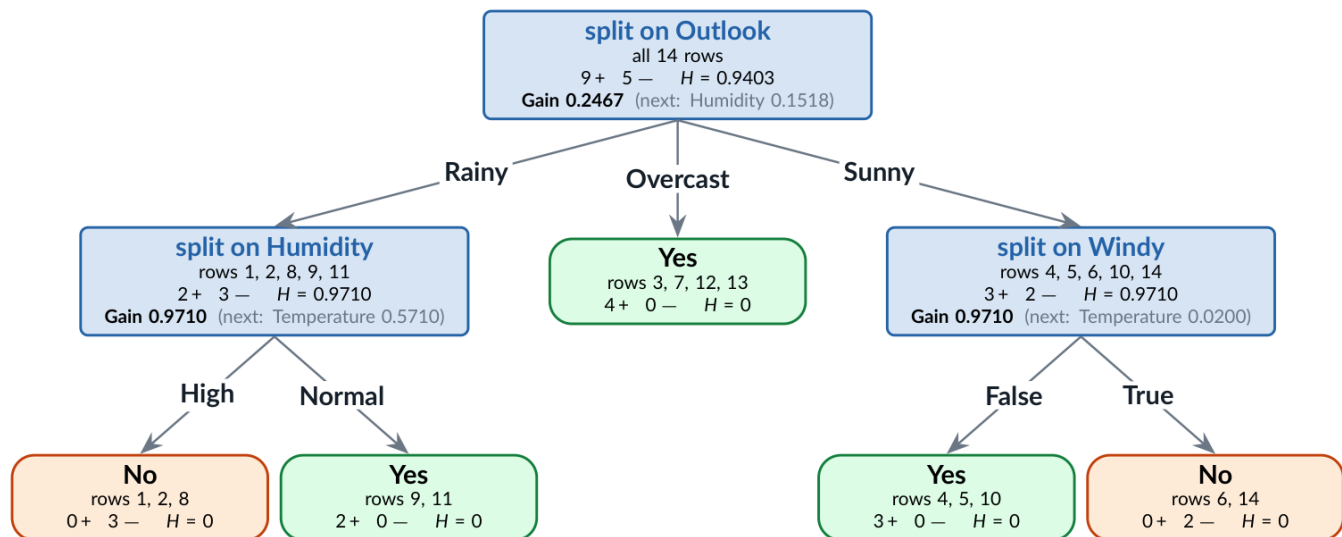
Outlook = Sunny rows 4, 5, 6, 10, 14 3 Yes, 2 No

Attribute	Value (rows)	P	N	H	Gain
Windy	False (4,5,10)	3	0	0	0.9710
	True (6,14)	0	2	0	
Temperature	Mild (4,10,14)	2	1	0.9183	0.0200
	Cool (5,6)	1	1	1.0000	
Humidity	High (4,14)	1	1	1.0000	0.0200
	Normal (5,6,10)	2	1	0.9183	

Windy: residual 0, gain 0.9710. Temperature and Humidity each leave one 1:1 pair and one 2:1 triple: residual $\frac{2}{5}(1) + \frac{3}{5}(0.9183) = 0.9510$, gain 0.0200. **Split on Windy.**

The tree, three levels and five leaves, one rule per leaf: Outlook = Overcast → Yes; Outlook = Rainy, Humidity = High → No; Outlook = Rainy, Humidity = Normal → Yes; Outlook = Sunny, Windy = False → Yes; Outlook = Sunny, Windy = True → No. Temperature never appears: **ID3 has decided it is irrelevant**, which is the feature-selection side of the algorithm.

The finished tree: play golf (rows numbered 1 to 14 top to bottom)



6.N.3 Problem 1.2 – the “play cricket” table, whose answer is different

How is the best attribute selected in a decision tree? Select the root attribute from the given sample data 1+7

79 Jth 1+7 · four rows differ from Problem 1.1 and the answer changes · rows 1 and 11 carry a different label, row 8 a different Windy value, and the target is renamed *Play cricket*

QUESTION AS PRINTED 79 Jth · Q8b · 1+7

How best attribute is selected in a decision-tree? Select the root attribute of the decision-tree from given sample data.

Outlook	Temperature	Humidity	Windy	Play cricket
Rainy	Hot	High	False	Yes
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	True	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	No
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Step 1 Entropy of the whole set

Still 9 Yes and 5 No out of 14, so $H(S) = 0.9403$ again – which is exactly why the table looks identical.

Steps 2 and 3

Attribute	Value	P	N	$ S_v $	$H(S_v)$	$\frac{ S_v }{14} H(S_v)$
Windy	False	7	0	7	0	0
	True	2	5	7	0.8631	$\frac{7}{14}(0.8631) = 0.4316$
residual = 0.4316						Gain = $0.9403 - 0.4316 = 0.5087$
Outlook	Rainy	2	3	5	0.9710	0.3468
	Overcast	4	0	4	0	0
	Sunny	3	2	5	0.9710	0.3468
residual = 0.6935						Gain = 0.2467
Temperature	Hot	3	1	4	0.8113	0.2318
	Mild	3	3	6	1.0000	0.4286
	Cool	3	1	4	0.8113	0.2318
residual = 0.8922						Gain = 0.0481
Humidity	High	4	3	7	0.9852	0.4926
	Normal	5	2	7	0.8631	0.4316
residual = 0.9242						Gain = 0.0161

Answer. The root attribute is **Windy**, gain **0.5087** – more than twice Outlook's 0.2467.

- **Compare with Problem 1.1, where Windy was the second worst attribute at 0.0481.** Three changed labels turned it into the best one.
- **Why:** in this table **every one of the seven non-windy days is a Yes**. One branch of Windy is completely pure and holds half the data, which is the largest single reduction in uncertainty available.
- **The lesson worth writing in the answer:** never carry a remembered root across from a similar-looking table. Recompute.

Finishing the tree (it is only two levels deep)

- **Windy = False** – rows 1, 3, 4, 5, 9, 10, 13, all Yes. **Leaf: Yes.**
- **Windy = True** – rows 2, 6, 7, 8, 11, 12, 14: 2 Yes 5 No, $H = 0.8631$. Steps 1 to 4 again on those seven rows, with the three attributes left:

Attribute	Value (rows)	P	N	H	Gain
Outlook	Rainy (2,8,11)	0	3	0	0.8631
	Sunny (6,14)	0	2	0	
	Overcast (7,12)	2	0	0	
Temperature	Hot (2)	0	1	0	0.1138
	Cool (6,7)	1	1	1.0000	
	Mild (8,11,12,14)	1	3	0.8113	
Humidity	High (2,8,12,14)	1	3	0.8113	0.0060
	Normal (6,7,11)	1	2	0.9183	

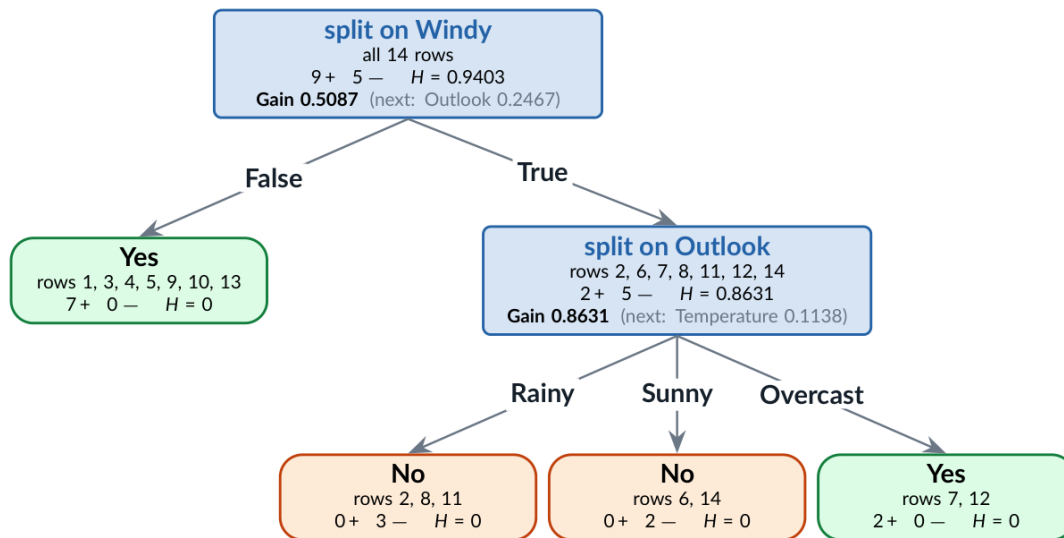
Outlook: every branch pure, residual 0, gain = 0.8631. Temperature: residual

$\frac{2}{7}(1) + \frac{4}{7}(0.8113) = 0.7493$, gain 0.1138. Humidity: residual

$\frac{4}{7}(0.8113) + \frac{3}{7}(0.9183) = 0.8571$, gain 0.0060. **Split on Outlook.**

- **The tree**, four leaves and only two tests: Windy = False -> Yes; Windy = True, Outlook = Rainy -> No; Windy = True, Outlook = Windy = True, Outlook = Overcast -> Yes.
- Neither Temperature nor Humidity is ever tested. The cricket tree is **smaller than the golf tree** – two tests against three – which is what a higher root gain buys you.

The finished tree: play cricket, two tests instead of three



6.N.4 Problem 1.3 – the mushroom island

Which attribute should you choose as the root of a decision tree? Give reason 8

77 Ch 8 · the attributes are already 0/1, so no entropy is wasted on multi-way splits · **only A to H are training data**; U, V and W carry ? in the Edible column and are the cases to be classified afterwards

QUESTION AS PRINTED 77 Ch · Q7 · 8

You are locked on a deserted island. Mushrooms of various types grow widely all over the island but no other food is anywhere to be found. Some of the mushrooms have been determined as poisonous and others as not (Determined by your former companions' trial and error). You are the only one reaming on the island. You have the following data to consider;

Example	NotHeavy	Smelly	Spotted	Smooth	Edible
A	1	0	0	0	1
B	1	0	1	0	1
C	0	1	0	1	1
D	0	0	0	1	0
E	1	1	1	0	0
F	1	0	1	1	0
G	1	0	0	1	0
H	0	1	0	0	0
U	0	1	1	1	?
V	1	1	0	1	?
W	1	1	0	0	?

You know whether or not mushroom A through H are poisonous, but you do not know about U through W. Which attributes should you choose as the root of a decision tree? Give reason.

The paper's own wording, slips included: "the only one reaming" for *the only one remaining*.

Step 1 Entropy of the training set

Rows A-H only. Edible = 1 for A, B, C → P = 3; Edible = 0 for D, E, F, G, H → N = 5.

$$H(S) = -\frac{3}{8} \log_2 \frac{3}{8} - \frac{5}{8} \log_2 \frac{5}{8} = 0.5306 + 0.4238 = \mathbf{0.9544}$$

Steps 2 and 3 All four attributes

Attribute	Value (rows)	P	N	$ S_v $	$H(S_v)$	$\frac{ S_v }{8} H(S_v)$
Smooth	1 (C,D,F,G)	1	3	4	0.8113	0.4056
	0 (A,B,E,H)	2	2	4	1.0000	0.5000
residual = 0.9056						Gain = 0.0488
NotHeavy	1 (A,B,E,F,G)	2	3	5	0.9710	0.6069
	0 (C,D,H)	1	2	3	0.9183	0.3444
residual = 0.9512						Gain = 0.0032
Smelly	1 (C,E,H)	1	2	3	0.9183	0.3444
	0 (A,B,D,F,G)	2	3	5	0.9710	0.6069
residual = 0.9512						Gain = 0.0032
Spotted	1 (B,E,F)	1	2	3	0.9183	0.3444
	0 (A,C,D,G,H)	2	3	5	0.9710	0.6069
residual = 0.9512						Gain = 0.0032

Answer. Choose **Smooth** as the root: gain **0.0488**, fifteen times the gain of any other attribute.

The reason the question asks for

- Smooth is the only attribute that produces an **uneven** class split: of the four smooth mushrooms only one is edible (3:1), while the four non-smooth ones are an even 2:2.
- The other three attributes split the data identically** – each one puts 1 edible and 2 inedible on one side and 2 edible and 3 inedible on the other, so all three have exactly the same entropy and exactly the same gain, 0.0032. Say this: it is the observation the examiner is looking for, and it means the choice between those three is arbitrary.

Finishing the tree (not asked; it is how U, V and W get classified)

Each branch uses only its own four mushrooms and only the three attributes Smooth has not used.

Smooth = 0 A, B, E, H 2 edible, 2 not, $H = 1$

Attribute	Value (rows)	P	N	H	Gain
Smelly	0 (A,B)	2	0	0	1.0000
	1 (E,H)	0	2	0	
NotHeavy	1 (A,B,E)	2	1	0.9183	0.3113
	0 (H)	0	1	0	
Spotted	1 (B,E)	1	1	1.0000	0
	0 (A,H)	1	1	1.0000	

NotHeavy: residual $\frac{3}{4}(0.9183) = 0.6887$, gain 0.3113. Spotted leaves both halves 1:1, gain 0. **Split on Smelly.**

The tree: Smooth = 0, Smelly = 0 → Edible; Smooth = 0, Smelly = 1 → Poisonous; Smooth = 1, Smelly = 1 → Edible; Smooth = 1, Smelly = 0 → Poisonous. Smelly flips meaning between the two branches, which is why it looked useless on its own at the root.

- U: Smooth = 1, Smelly = 1 → predict **Edible**.
- V: Smooth = 1, Smelly = 1 → predict **Edible**.
- W: Smooth = 0, Smelly = 1 → predict **Poisonous**.
- The honest caveat:** the leaf U and V land in holds a single training mushroom, C. Eight examples cannot make that leaf trustworthy, so the answer should say the tree predicts *edible* on the evidence of one mushroom – and on an island where the cost of a wrong guess is death, that is a reason to abstain.

Two things worth adding for the last marks

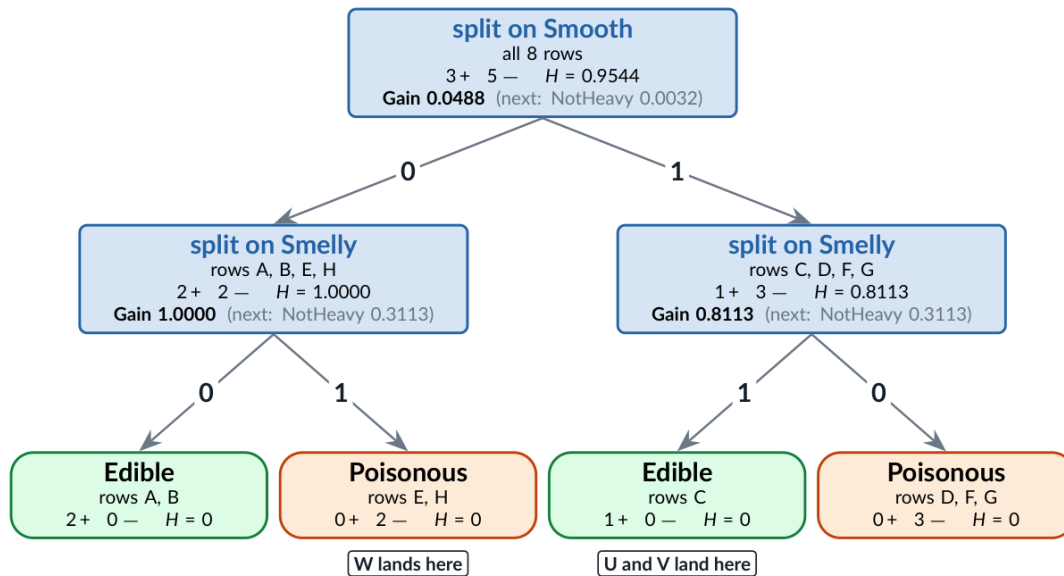
- No attribute here is much use at the root.** The best gain is 0.0488 against a starting entropy of 0.9544, so the root removes only about 5 % of the uncertainty.
- But the second level is perfect.** Under either value of Smooth, **Smelly** separates the mushrooms completely (tables below). The tree is therefore only two tests deep, and it classifies U, V and W: the story implies that even though the question stops at the root.
- Beware the root-only shortcut: reading U and V off the 3:1 Smooth branch says *not edible*, and the full tree says the opposite.

Smooth = 1 C, D, F, G 1 edible, 3 not, $H = 0.8113$

Attribute	Value (rows)	P	N	H	Gain
Smelly	1 (C)	1	0	0	0.8113
	0 (D,F,G)	0	3	0	
NotHeavy	0 (C,D)	1	1	1.0000	0.3113
	1 (F,G)	0	2	0	
Spotted	0 (C,D,G)	1	2	0.9183	0.1226
	1 (F)	0	1	0	

NotHeavy: residual $\frac{2}{4}(1) = 0.5000$, gain 0.3113. Spotted: residual $\frac{3}{4}(0.9183) = 0.6887$, gain 0.1226. **Split on Smelly.**

The finished tree: the mushroom island, with U, V and W dropped through it



6.N.5 Problem 1.4 – the profit table (extra practice, not a PYQ)

PROBLEM not a PYQ · *Insights* Example 6.4, the worked example the textbook uses

Build a decision tree for the following data using ID3.

Age	Competition	Type	Profit
Old	Yes	Software	Down
Old	No	Software	Down
Old	No	Hardware	Down
Mid	Yes	Software	Down
Mid	Yes	Hardware	Down
Mid	No	Hardware	Up
Mid	No	Software	Up
New	Yes	Software	Up
New	No	Hardware	Up
New	No	Software	Up

Level 1

10 rows, 5 Up and 5 Down, so $H(S) = 1.0000$ exactly – the easiest possible starting point.

Attribute	Value	P	N	$ S_v $	$H(S_v)$	$\frac{ S_v }{10}H$
Age	Old	0	3	3	0	0
	Mid	2	2	4	1.0000	0.4000
	New	3	0	3	0	0
residual = 0.4000						Gain = 0.6000
Competition	Yes	1	3	4	0.8113	0.3245
	No	4	2	6	0.9183	0.5510
residual = 0.8755						Gain = 0.1245
Type	Software	3	3	6	1.0000	0.6000
	Hardware	2	2	4	1.0000	0.4000
residual = 1.0000						Gain = 0

- **Type has gain exactly zero:** both its branches are split 50/50, the same as the parent, so it tells you nothing at all. That is the cleanest illustration in this chapter of what a useless attribute looks like.
- Root = **Age**. Old → Down, New → Up, both pure. Only *Mid* needs a second level.

Level 2, the Mid branch

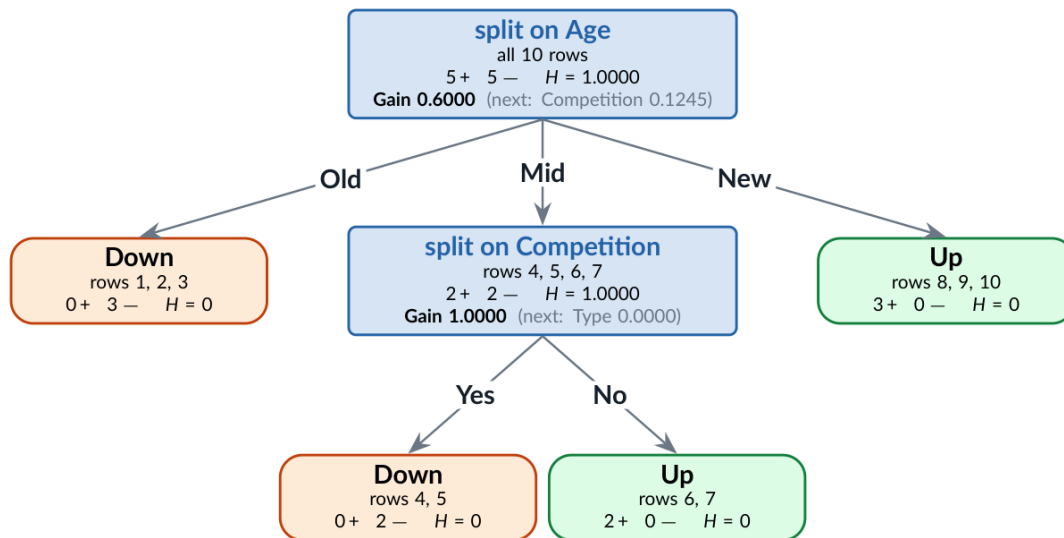
4 rows, 2 Up and 2 Down, $H = 1$.

Attribute	Value	P	N	H	Gain
Competition	Yes	0	2	0	
	No	2	0	0	1.0000
Type	Software	1	1	1	
	Hardware	1	1	1	0.0000

Answer. Age = Old → Down;
 Age = New → Up;
 Age = Mid, Competition = Yes → Down;
 Age = Mid, Competition = No → Up.

- Two tests, four leaves, and Type is never used – the tree *Insights* prints, and its arithmetic (0.6, 0.13, 0) checks out to the second decimal.
- **The general shape to remember:** an attribute whose branches are all pure has residual 0 and therefore gain $= H(S)$, the maximum possible. Whenever you see one, stop comparing and take it.

The finished tree: profit (rows numbered 1 to 10 top to bottom)



6.N.6 Genetic algorithms: the method, then three runs

The recipe every GA problem follows

- 1. **Encode.** Say what one chromosome means, gene by gene. Almost every exam problem is a binary string with one bit per yes/no decision, or a string of digits.
- 2. **Initialise.** Take the population the question gives you. If it gives none, write out four or five random strings and say they are random.
- 3. **Evaluate.** Apply the fitness function to every chromosome, showing the substitution. **An infeasible chromosome scores 0**, and saying *why* it is infeasible earns the mark.
- 4. **Select.** Rank them; keep the best, drop the worst.
- 5. **Crossover.** State the point, draw the cut with a bar, write both offspring, and **evaluate them too**.
- 6. **Mutate.** Flip one gene, say which, and evaluate.
- 7. **Form the next generation** and compare its best and mean fitness against the previous one. **That comparison is the answer**; a GA solution without it is only half a solution.

6.N.7 Problem 2.1 – the knapsack, one full generation

PROBLEM not a PYQ · *Insights* Example 6.1, the standard GA demonstration

A thief has a knapsack of limited capacity that can hold almost 12 kg. Which items should be kept in the knapsack so that its value is maximized? Solve using a genetic algorithm.

Item	Weight	Value
A	5 kg	\$12
B	3 kg	\$5
C	7 kg	\$10
D	2 kg	\$7

Step 1 Chromosome encoding

- Four items, so a **4-bit chromosome** in the fixed order A B C D.
- Gene = 1 means the item is in the knapsack, 0 means it is not. The search space is $2^4 = 16$ chromosomes.

Step 2 Initial population and fitness

Fitness = total value if total weight ≤ 12 kg, otherwise 0.

Chromo	Items	Weight	Value	Fitness
C_1 0010	C	7 kg	10	10
C_2 0110	B, C	10 kg	15	15
C_3 1001	A, D	7 kg	19	19
C_4 1111	A,B,C,D	17 kg	34	0

- C_4 is the heaviest *and* the most valuable, and it scores **zero**: 17 kg will not fit in a 12 kg sack, so it is not a solution at all.
- Generation 1: best = 19, mean = $\frac{10+15+19+0}{4} = 11.0$.

Step 6 Generation 2

Chromosome	Weight	Value	Fitness
1101 (mutant)	10 kg	24	24
1010 (O_1)	12 kg	22	22
1001 (C_3)	7 kg	19	19
0110 (C_2)	10 kg	15	15

- Best 19 $\rightarrow$ **24**; mean 11.0 $\rightarrow$ **20.0**. Both rose, which is what you must state.
- The two weakest of generation 1 (C_1 and the infeasible C_4) have been replaced by the offspring and the mutant.

Step 3 Selection

- Rank: C_3 (19), C_2 (15), C_1 (10), C_4 (0).
- C_4 has zero fitness so a roulette wheel gives it **no slice at all** and it never reproduces. Take C_3 and C_2 as parents.

Step 4 Single-point crossover after gene 2

	String	Weight	Fitness
Parent C_3	1 0 0 1	7 kg	19
Parent C_2	0 1 1 0	10 kg	15
Offspring O_1	1 0 1 0	12 kg	22
Offspring O_2	0 1 0 1	5 kg	12

- $O_1 = 1010$ is A and C: $5 + 7 = 12$ kg, exactly the capacity, value $12 + 10 = 22$. **Better than either parent**, which is crossover doing its job.

Step 5 Mutation

- Flip **gene 2** of $C_3 = 1001$: $1001 \rightarrow 1101$.
- 1101 is A, B and D: $5 + 3 + 2 = 10$ kg, value $12 + 5 + 7 = 24$.

Answer. Take A, B and D: 10 kg, value \$24. Chromosome 1101.

Termination, and the point of the whole problem

- An exhaustive check of all 16 chromosomes confirms 24 is the **global optimum**; the next best feasible sacks are 1010 and 0111, both worth 22. So the run stops here, having converged.
- Notice which operator found it.** Crossover took 19 to 22. It could never have reached 24, because *no chromosome in the population carried a 1 in gene B and gene D together* – crossover can only shuffle genes that already exist. Mutation supplied the missing bit.
- That is the complete answer to 81 Ba's "justify why mutation is important", and it is far stronger than quoting the definition.

6.N.8 Problem 2.2 – fitness, ranking and crossover on eight-gene chromosomes**Rank the chromosomes by fitness, then perform the crossover operations** 2+8

75 Bh 2+8 · the paper prints no chromosome list – the sentence ends and Q7 follows · the four chromosomes below are the ones the standard version of this question carries (Insights Example 6.2) · the swarm-intelligence half of the question is answered in §6.9 of the theory

QUESTION AS PRINTED 75 Bh · Q6 · 2+8

What do you understand by swarm intelligence? Suppose chromosomes are of the form $x = a b c d e f g h$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as:

$$f(x) = (a + b) - (c + d) + (e + f) - (g + h)$$

and let the initial population consist of four individuals with the following chromosomes.

DEFECT IN THE PAPER: the sentence ends there and no chromosome list is printed. Confirmed against the scan. Worked below with the standard four.

SUPPLIED the four chromosomes the textbook version of this question prints

$$x_1 = 65413532 \quad x_2 = 87126601 \quad x_3 = 23921285 \quad x_4 = 41852094$$

(a) Evaluate the fitness of each, and rank them fittest first. (b) Perform single-point crossover of the two fittest at the middle point, and a two-point crossover of x_1 and x_3 .

(a) Step 1 Fitness of each chromosomeRead the eight digits as a to h in order, then substitute.

	Chromosome	Substitution	$f(x)$
x_1	6 5 4 1 3 5 3 2	$(6+5) - (4+1) +$ $(3+5) - (3+2)$ $= 11 - 5 + 8 - 5$	9
x_2	8 7 1 2 6 6 0 1	$(8+7) - (1+2) +$ $(6+6) - (0+1)$ $= 15 - 3 + 12 - 1$	23
x_3	2 3 9 2 1 2 8 5	$(2+3) - (9+2) +$ $(1+2) - (8+5)$ $= 5 - 11 + 3 - 13$	-16
x_4	4 1 8 5 2 0 9 4	$(4+1) - (8+5) +$ $(2+0) - (9+4)$ $= 5 - 13 + 2 - 13$	-19

Step 2 Rank

Answer (a). Fittest first: x_2 (23), x_1 (9), x_3 (-16), x_4 (-19).
 Mean fitness of generation 1 = $\frac{9+23-16-19}{4} = -0.75$.

- **Read the fitness function before you compute:** genes a, b, e, f are *added* and c, d, g, h are *subtracted*, so a fit chromosome is large in positions 1, 2, 5, 6 and small in 3, 4, 7, 8. x_2 is exactly that shape; x_4 is its mirror image, and scores -19.
- The theoretical maximum is $f = 36$ at 99009900, the minimum -36.

(b) Step 3 Single-point crossover of the two fittest, at the middle

Eight genes, so the middle point is after gene 4.

	String	Substituti	f
x_2	8712 6601		23
x_1	6541 3532		9
O_1	8712 3532	$15 - 3 +$ $8 - 5$	15
O_2	6541 6601	$11 - 5 +$ $12 - 1$	17

- Both offspring beat x_1 ; neither beats x_2 . **Mean rises from -0.75 to 16** if the two weakest are replaced by O_1 and O_2 .

Step 4 Two-point crossover of x_1 and x_3 , cuts after genes 2 and 6

	String	Substitu	f
x_1	65 4135 32		9
x_3	23 9212 85		-16
O_1	65 9212 32	$11 -$ $11 +$ $3 - 5$	-2
O_2	23 4135 85	$5 - 5 +$ $8 - 13$	-5

- In a two-point crossover the **middle segment** is exchanged and the two outer segments stay put.
- **Both offspring are worse than x_1** , and that is worth stating: crossover does not guarantee improvement. It is *selection*, applied next generation, that throws these away and keeps the gains.

6.N.9 Problem 2.3 – reaching a target fitness, and a textbook crossover that does nothing**PROBLEM** not a PYQ · *Insights* Example 6.3, reworked – see the note

You are given a coin. Find the generations using a genetic algorithm so that you can get the number of heads above 30. Take 5 chromosomes made up of 10 genes to proceed with the solution.

Steps 1 and 2 Encoding and the initial population

- Ten tosses per chromosome: gene 1 = **head**, 0 = **tail**.
- Fitness of a chromosome = its number of heads. The *objective* is on the population: **total heads** > 30 across all five.

	Chromosome	f = heads
C_1	1 0 0 0 1 0 1 1 0 0	4
C_2	1 1 1 0 0 1 0 1 1 0	6
C_3	1 0 1 0 1 1 0 0 1 0	5
C_4	0 1 1 1 1 0 1 0 1 1	7
C_5	0 1 1 1 0 0 1 1 1 1	7
Total		29 < 30, so continue

Step 3 Selection

- The two fittest are C_4 and C_5 , both 7. Take them as parents.

Step 4 Choosing the crossover point matters here

- C_4 and C_5 **share their first four genes**, 0111. Cut anywhere in the first four positions and the “offspring” are just the two parents swapped – no new chromosome exists and the population cannot improve.
- So cut **after gene 6**, where the parents actually differ.

	String	f
C_4	011110 1011	7
C_5	011100 1111	7
O_1	011110 1111	8
O_2	011100 1011	6

Step 5 Generation 2: keep the best five of the seven

Chromosome	f	
O_1 0111101111	8	new best
C_4 0111101011	7	
C_5 0111001111	7	
C_2 1110010110	6	
O_2 0111001011	6	
Total	34	

Answer. Total heads 29 → **34**, and the best individual 7 → **8**. The objective is reached in the **second generation**.

Where the textbook goes wrong, and why it matters

- *Insights* crosses the same two parents *inside* the four genes they share, so its O_1 and O_2 come out equal to C_5 and C_4 . Its total still reaches 34, but only because the best chromosome has been **counted twice**; the best individual never improves past 7.
- Cutting where the parents differ produces a genuinely new chromosome with 8 heads. **An examiner reading the two solutions can see which one understood crossover.**

6.N.10 Problem 3.1 – fuzzy set operations

PROBLEM not a PYQ directly, but the arithmetic behind every “explain fuzzy operations” answer

For the fuzzy sets $A = \{(10, 0.2), (20, 0.3), (30, 0.4), (40, 0.5)\}$ and $B = \{(10, 0.3), (20, 0.4), (30, 0.1), (40, 0.2)\}$ over the universe $U = \{10, 20, 30, 40\}$, find $A \cup B$, $A \cap B$, A' , the bold union $A \oplus B$ and the bold intersection $A \odot B$.

Work element by element

Every fuzzy operation is defined *pointwise*: apply it to the membership values at each x independently. There is nothing else to it.

x	10	20	30	40
$\mu_A(x)$	0.2	0.3	0.4	0.5
$\mu_B(x)$	0.3	0.4	0.1	0.2
$A \cup B = \max$	0.3	0.4	0.4	0.5
$A \cap B = \min$	0.2	0.3	0.1	0.2
$A' = 1 - \mu_A$	0.8	0.7	0.6	0.5
$A \oplus B = \min(1, \mu_A + \mu_B)$	0.5	0.7	0.5	0.7
$A \odot B = \max(0, \mu_A + \mu_B - 1)$	0	0	0	0

- **Bold intersection is empty** because no pair of memberships sums past 1. Say that; it is the check that you applied the formula rather than guessing.
- **Note** $A \cap A' \neq \emptyset$: at $x = 40$ both are 0.5, so their intersection is 0.5. In crisp set theory that is impossible. This one line is the sharpest possible answer to “how does fuzzy logic differ from classical logic”, and the same goes for $A \cup A' \neq U$.

Answers.

A	$\cup$	B	=
$\{(10, 0.3), (20, 0.4), (30, 0.4), (40, 0.5)\}$			
A	$\cap$	B	=
$\{(10, 0.2), (20, 0.3), (30, 0.1), (40, 0.2)\}$			
$A' = \{(10, 0.8), (20, 0.7), (30, 0.6), (40, 0.5)\}$			
A	$\oplus$	B	=
$\{(10, 0.5), (20, 0.7), (30, 0.5), (40, 0.7)\}$			
$A \odot B = \{(10, 0), (20, 0), (30, 0), (40, 0)\}$			

One correction to the textbook

- *Insights* p167 prints the complement as $A' = \{(10, 0.8), (20, 0.7), (30, \mathbf{0.5}), (40, 0.5)\}$.
- $\mu_A(30) = 0.4$, so $1 - 0.4 = \mathbf{0.6}$, not 0.5. Every other value in that box is right; only this one is a slip.
- The check that catches it in one second: $\mu_A(x) + \mu_{A'}(x)$ **must equal exactly 1 at every element**. Run it down the column before you hand the paper in.

6.N.11 Problem 3.2 — fuzzification: turning a crisp reading into a membership

PROBLEM not a PYQ · *Insights* p166, the step every fuzzy question needs

The linguistic term *medium speed* has a membership function that rises linearly from 0 at 7.5 km/h to 1 at 12.5 km/h. Fuzzify the readings 10 km/h and 9 km/h.

The one formula

On a rising limb from a to b :

$$\mu(x) = \frac{x - a}{b - a} \quad a \leq x \leq b$$

- **At 10 km/h:** $\mu = \frac{10 - 7.5}{12.5 - 7.5} = \frac{2.5}{5} = 0.5$, so the fuzzy pair is (10, 0.5).
- **At 9 km/h:** $\mu = \frac{9 - 7.5}{5} = \frac{1.5}{5} = 0.3$, so the pair is (9, 0.3).
- On a *falling* limb from b to c the formula is $\mu(x) = \frac{c - x}{c - b}$, and a triangle is one of each.

Two things this small problem teaches

- **The first number in the pair is the element, not the peak.** *Insights* writes the second answer as (10, 0.3), but the reading was 9 km/h, so the pair is (9, 0.3). The membership value is right; the label is not.
- A crisp reading normally fuzzifies to **more than one term at once**, because the membership functions overlap: 10 km/h might be *slow* to degree 0.5 and *medium* to degree 0.5 at the same time. That overlap is the whole reason fuzzy control is smooth, so never report only one term unless the reading really is outside every other set.

6.N.12 Problem 3.3 — a complete Mamdani fuzzy inference

Which fuzzy answer does this paper want? Sixteen papers ask something about fuzzy logic and they do not want the same answer. Decide from *what is asked for*, not from the word *fuzzy*. “explain fuzzy inference / the Mamdani method / the steps, with an example”, 5 marks or more → §6.N.12, the worked run below. “the architecture”, or “the major steps for developing a system based on fuzzy logic” → §6.N.13: those ask you to *build* the variables, membership functions and rule base that §6.N.12 is handed for free. A 4-mark short note, or a 2-mark “what is fuzzy logic” → neither of them. That is Ch 6 §6.7, and running a defuzzification for it spends twenty minutes earning four marks.

Explain fuzzy inference / the Mamdani method / the steps of a fuzzy inference system, with an example 1+2+5 2+6 3+7

8

HOT 6/41 73 Ma 3+7, 80 Bh 8, 79 Bh 2+6, 70 Ma 2+6, 80 Ash 1+2+5, 79 Ash 1+2+5 · the papers never supply numbers, so **you have to invent the example** — this one is small enough to write out in full in about ten minutes · 73 Ma is the only paper in the 41 that names **Mamdani**; the other five say only *fuzzy inference* or *the steps*, and this one run answers all six

QUESTION AS PRINTED 73 Ma · Q7 · 3+7

What is Fuzzy Logic and why is it important? Explain the Mamdani Fuzzy Inference Method with example.

PROBLEM the worked example, chosen so every number is checkable

A fan-speed controller has two inputs, temperature T (0–40 °C) and humidity H (0–100 %), and one output, fan speed S (0–100 %).

Membership functions. *Cold*: 1 up to 10, falling to 0 at 20. *Warm*: triangle 10–20–30. *Hot*: 0 at 20, rising to 1 at 30 and staying there. *Low humidity*: 1 up to 30, falling to 0 at 60. *High humidity*: 0 at 30, rising to 1 at 60. Output sets *Slow*, *Medium*, *Fast* are triangles peaking at 0, 50 and 100.

Rules. R1: IF T is Cold AND H is Low THEN S is Slow. R2: IF T is Warm THEN S is Medium. R3: IF T is Hot OR H is High THEN S is Fast.

Find the fan speed for $T = 26$ °C and $H = 45$ %.

Step 1 Fuzzify both inputs

Read each crisp value off every membership curve it touches.

Term **Working at** $T = 26$, μ
 $H = 45$

Cold $26 > 20$, outside the set 0

Warm falling limb: $\frac{30-26}{30-20}$ **0.4**

Hot rising limb: $\frac{26-20}{30-20}$ **0.6**

Low falling limb: $\frac{60-45}{60-30}$ 0.5

High rising limb: $\frac{45-30}{60-30}$ 0.5

– 26 °C is **both warm and hot**, and 45 % is equally low and high. That is the point of fuzzy logic: no reading has to pick a side.

– Check: $\mu_{Warm} + \mu_{Hot} = 1$ and $\mu_{Low} + \mu_{High} = 1$, as they must be where exactly two curves overlap linearly.

Step 2 Fire the rules (AND = min, OR = max)

Rule **Strength** w

R1 $\min(\mu_{Cold}, \mu_{Low}) = \min(0, 0.5)$ **0**

R2 μ_{Warm} 0.4

R3 $\max(\mu_{Hot}, \mu_{High}) = \max(0.6, 0.5)$ 0.6

– **R1 does not fire at all** ($w = 0$), so *Slow* contributes nothing. Saying so explicitly is worth a mark.

Answer. Run the fan at **about 62 %**. Rules R2 and R3 fired at 0.4 and 0.6; R1 did not fire.

The six Mamdani operations, named against the steps above

- **1. Determine the rules** – R1, R2, R3, given.
- **2. Fuzzify the inputs** – Step 1.
- **3. Combine to get rule strength** – Step 2, fuzzy AND = min, OR = max.
- **4. Implication** – Step 3, clip each consequent set at its strength.
- **5. Aggregation** – Step 4, union the clipped sets with max.
- **6. Defuzzification** – Step 5, centre of gravity.

Step 3 Implication: clip each output set at its rule strength

– *Medium* is clipped at **0.4**, *Fast* at **0.6**. Mamdani implication is the **min** operator, so the top of each triangle is simply cut flat.

Step 4 Aggregation: take the max of the clipped sets

Sample the aggregated set every 10 units of S .

S	0	10	20	30	40	50
clipped	0	0.2	0.4	0.4	0.4	0.4
Medium						
clipped Fast	0	0	0	0	0	0
aggregate	0	0.2	0.4	0.4	0.4	0.4
S	60	70	80	90	100	
clipped	0.4	0.4	0.4	0.2	0	
Medium						
clipped Fast	0.2	0.4	0.6	0.6	0.6	
aggregate	0.4	0.4	0.6	0.6	0.6	

Step 5 Defuzzify by the centre of gravity

$$S^* = \frac{\sum S_i \mu(S_i)}{\sum \mu(S_i)} = \frac{272}{4.4} = \mathbf{61.8\%}$$

$$\sum \mu = 0 + 0.2 + 0.4 + 0.4 + 0.4 + 0.4 + 0.4 + 0.4 + 0.6 + 0.6 + 0.6 = 4.4;$$

$$\sum S\mu = 2 + 8 + 12 + 16 + 20 + 24 + 28 + 48 + 54 + 60 = 272.$$

Integrating the true area instead of sampling gives 58.8 %, so the ten-point sum is accurate to about 3 %, which is well inside what any examiner expects.

The shortcut, and when it is safe

– If you are short of time, the **weighted average of the consequent peaks** is one line:

$$S^* = \frac{0.4(50) + 0.6(100)}{0.4 + 0.6} = \frac{80}{1.0} = 80\%$$

- **It is not the same answer**, because it ignores the shape and the overlap of the two output sets. Use it only when the question says *weighted average* or *centre of sums*; if it says **centroid** or **centre of gravity**, do the sampled sum above.
- **Two checks before you write the final number**: the answer must lie *inside* the output range (0–100 here), and it must sit between the peaks of the sets that fired (50 and 100). Both 61.8 and 80 pass; a number outside that band means an arithmetic slip.

6.N.13 Problem 3.4 – designing the fuzzy system that Problem 3.3 runs

What are the major steps for developing a system based on fuzzy logic? Explain the architecture and working mechanism of a fuzzy inference system 2+5 2+6

PIN 2/41 80 Ch 2+5, 82 Ba 2+6 · **these two do not want a defuzzification**: §6.N.12 is handed its membership functions and its three rules, and these papers ask *where those came from* · the 2-mark opener, *when do we use fuzzy logic*, is the list in Ch 6 §6.7

QUESTION AS PRINTED 80 Ch · Q7 · 2+5

When shall we use Fuzzy Logic? Explain with an appropriate example. What are the major steps for developing a system based on Fuzzy logic?

QUESTION AS PRINTED 82 Ba · Q7 · 2+6

When do we use fuzzy logic? Explain the architecture and working mechanism of a fuzzy inference system.

The seven design steps, in the order you do them

- 1. Confirm fuzzy logic fits.** Expert rules exist, no exact plant model does, and the boundaries are linguistic. If a threshold already works, do not fuzzify it.
- 2. Name the variables and their universes of discourse.** One per sensor and one per actuator, each with the range it physically spans.
- 3. Choose the linguistic terms for each variable.** Three to five per variable; **the rule count is the product**, so five terms on three inputs is 125 rules and unmaintainable.
- 4. Draw one membership function per term.** Triangles and trapezoids, and **they must overlap**: every point of the universe belongs to at least one set, and neighbours cross at about 0.5.
- 5. Build the rule base** as a matrix of every input-term combination against the action to take. The domain expert fills the cells, not the engineer.
- 6. Fix the operators:** AND = min, OR = max, implication = min (Mamdani), aggregation = max, and pick the defuzzifier (centroid unless the question says otherwise).
- 7. Simulate, tune, iterate.** Run known cases, move the membership breakpoints until the outputs match the expert's judgement, then deploy.

Steps 1–5 are design and happen once. Steps 4 and 6 are what the running system executes on every reading — that is the *working mechanism* half of 82 Ba, and its block diagram is the architecture figure in §6.7.

The compression §6.N.12 uses is not a relabelling

- Whole rows and columns of that matrix agree, so it is tempting to write *Hot OR High* → *Fast* as one rule and be done. That is what §6.N.12's R3 does.
- It changes the numbers.** A matrix cell fires at $\min(\mu_T, \mu_H)$, so a fully high humidity is dragged down by a half-cold temperature. An OR rule takes max, and lets either antecedent carry the rule on its own.

At $T = 26, H = 45$	3 OR rules	6 cells
Slow	0	0
Medium	0.4	0.4
Fast	0.6	0.5
centroid	61.8 %	59.8 %

Worked step 2–4 for the fan controller of §6.N.12

Variable	Universe	Terms, and where they cross
T in	0–40 °C	Cold, Warm, Hot. Cold/Warm cross at 15 , Warm/Hot at 25 , both at $\mu = 0.5$
H in	0–100 %	Low, High, crossing at 45
S out	0–100 %	Slow, Medium, Fast, peaking at 0, 50, 100

- This is why §6.N.12 read $\mu_{Low} = \mu_{High} = 0.5$:** its 45 % is exactly the crossover you placed here. A reading that lands on a crossover is the best sanity check a design gets.

Step 5 The rule matrix, all six cells

	H Low	H High
T Cold	Slow	Fast
T Warm	Medium	Medium and Fast
T Hot	Fast	Fast

- The Warm/High cell naming two actions is not a bug.** Both fire, at their own strengths, and aggregation settles it. A crisp rule base would have to pick one and would jerk at the boundary.

- Over the whole input grid the two rule bases give different firing strengths at **1386 of the 4141** integer (T, H) points, and they are furthest apart at $T = 15, H = 60$: the OR rule fires Fast at **1.0** because the air is fully humid, the matrix at **0.5** because min sees a temperature that is only half cold.
- Which one to write:** either, but **say which**. The marks are for naming the operators and using them consistently, not for the crisp number — no paper in the 41 supplies data, so no paper has a right answer to compare against.
- Do not print the matrix and then solve with the OR rules without a word**, which is the one version that is simply inconsistent.

Answer. Seven steps: confirm the fit, name the variables and their ranges, choose linguistic terms, draw overlapping membership functions, fill the rule matrix, fix the operators and the defuzzifier, then simulate and tune. **Design ends where §6.N.12 begins.**

What loses marks here

- **Answering with a defuzzification.** Neither paper gives numbers, and neither asks for a crisp output. Five or six marks are for the *procedure*.
- **Membership functions that do not overlap.** Then the system is a lookup table with fuzzy notation, and it jerks at every boundary. Say the overlap is deliberate.
- **Forgetting the 2-mark opener.** Both papers open with *when do we use fuzzy logic*, and it is a separate list: vague boundaries, expert rules available, no mathematical model, tolerance for imprecision. **§6.7.**

Sources and verification

- All three PYQ data tables are transcribed from `AI/ocr/CT-653_CT-710_CT-78506_NewScan_OCR.md`, cell by cell.
- Method throughout follows *Insights on Artificial Intelligence* ch 6, checked independently. **Two of its numbers are wrong and are corrected here:** the complement in Problem 3.1 and the element label in Problem 3.2. **One of its methods is wrong:** the crossover in Problem 2.3.
- Every entropy, gain, fitness, membership and defuzzified value on these pages is recomputed from the problem statement by `verify.py`, which also proves the knapsack optimum by exhaustive search over all 16 chromosomes and confirms that *Insights'* crossover point produces offspring identical to its parents.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
AI	artificial intelligence	ANI	artificial narrow intelligence
AGI	artificial general intelligence	ASI	artificial super intelligence
KR	knowledge representation	KRR	knowledge representation and reasoning
PEAS	performance measure, environment, actuators, sensors	GPS	General Problem Solver
NLP	natural language processing	NLU	natural language understanding
NLG	natural language generation	CAPTCHA	completely automated public Turing test to tell computers and humans apart
ML	machine learning	ANN, NN	artificial neural network
MLP	multilayer perceptron	BP	back-propagation
SOM	self-organizing map	GA	genetic algorithm
FIS	fuzzy inference system	ID3	Iterative Dichotomiser 3
CSP	constraint satisfaction problem	DFS	depth first search
BFS	breadth first search	DLS	depth limited search
IDS	iterative deepening search	UCS	uniform cost search
CNF	conjunctive normal form	WFF	well-formed formula
FOL	first-order logic	FOPL	first-order predicate logic
RRS	resolution refutation system	BBN	Bayesian belief network
Short	Means	Short	Means
ES	expert system	KB	knowledge base
CD	conceptual dependency	LISP	list processing (the language)
MOP	memory organization packet	isa	the class-to-class link of a semantic net (<i>instance</i> is the individual-to-class one)
attr(s)	attribute(s)	repo	repository
freq	frequency	config	configuration
info	information	no.	number
Eg	example	Vs	compare / differentiate
w/, w/o	with, without	+Fig	draw the figure
→	leads to, therefore	=	is, equals

Reading the year tags and chips

Exam-paper month codes: Ba = Baishakh · Jth = Jestha · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Po = Poush · Ma = Magh · Ch = Chaitra. Years are Bikram Sambat, so 80 Ch is 2080 Chaitra.

Bold year tags are Regular exams, plain are Back exams. A header printing “Regular / Back” in one box counts as Regular.

Three course codes, three faces. Plain = **CT 653** (BCT, III-II, 26 papers) · `monospace` = CT 710 (BEI, IV-I, 7 papers) · *italic* = CT 78506 (BEX, IV-II, 8 papers). **The same year code names different papers under different codes:** 77 Ch and 77 Ch are two different exams, and only the font tells them apart.

Tier chips over all 41 papers, 2069–2082 BS: **TOP /41** = topic asked in **8 or more** papers · **HOT /41** = **4 to 7** papers · **PIN /41** = **1 to 3** papers but worth ≥ 6 marks.

Mark chips such as **2+6** are the marks that question actually carried. `syllabus, no PYQ yet` marks a syllabus topic never yet examined but written up anyway. `verified/added` marks anything researched and added that no source note in this folder covers.